

SYSTEMS.

The term **System** can be defined as a collection of independent entities that collectively work together to achieve a desired goal.

All things can be viewed as being made up of small independent components (subsystems) that come together to form a bigger more complex system.

For example;

- (1). A School can be seen as a system with students, teachers, Accounts department, and the Administration as subsystems. The school system itself is a subsystem of the ministry of education.

COMPUTER SYSTEMS.

What is a Computer system?

- ❖ The term **Computer system** refers to the complete set of devices required to use & operate the computer.
- ❖ Computer system is the complete set of devices that make a computer work as one unit.
- ❖ A collection of entities that work together to process and manage information using computers.

A computer system consists of the computer itself & supporting devices for input, output, processing & storage of data such as disks, Monitors, Printers, etc

MICROCOMPUTER.

This is a computer whose Central Processing unit (CPU)/ Processor has been implemented with a Microprocessor.

- It is a Desktop computer.
- It's made of very small, tiny gadgets (micro-chips), which have been developed as a result of miniaturization of technology.

A microcomputer uses one or more boards to implement all the functions of a complete system.

In small businesses, the microcomputer configuration typically includes;

- (i). A Keyboard and/or Mouse for input.
- (ii). A Winchester disk, usually 3½" disk unit, for reading & writing onto the 3½" diskettes.
- (iii). A slow Printer, usually a Character printer.

Uses of a Microcomputer.

- 1). Used in a modern office for:
 -) Word processing - for production & printing of all types of business documents.
 -) Accounting purposes & financial modeling.
 -) Record keeping & analysis.
 -) Desktop publishing.
- 2). Used for all types of communication (both external and internal) & bulletins.
- 3). Entertainment and all other types of voice output.
- 4). Used for project management & statistical analysis.
- 5). For information management, i.e., for storage of files & folders.
- 6). For Electronic learning.
- 7). Animation and simulation.

Microprocessor.

This is a small computer processor (usually excluding Main memory) manufactured on a single chip.

The Microprocessor is the brain (powerhouse) of the Personal Computer (PC). It is an electronic circuit that handles Input/Output signals from the peripheral devices of a computer.

It is a Very Large Scale Integration (VLSI) component that performs all the calculations and processing of a computer system.

Chip -The small rectangular piece of Silicon on which most ICs are implemented.

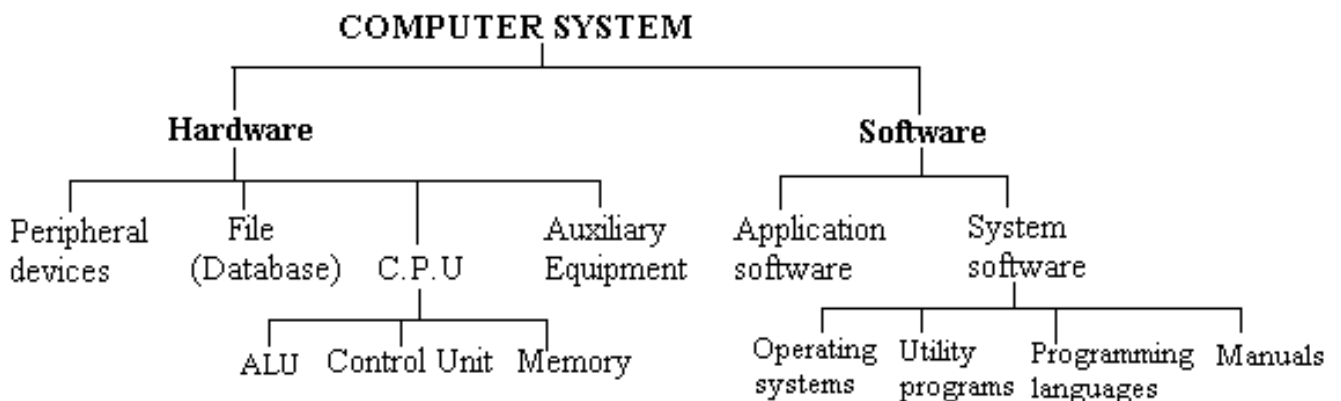
Microprocessor System.

Refers to the electronic boards required to implement a functional computer. Generally, they do not include the Chassis, Power supply unit or peripheral devices.

FUNCTIONAL ORGANIZATION OF THE ELEMENTS OF A COMPUTER SYSTEM.

A Computer system consists (or is made up) of 4 basic elements that are interrelated and work in unison. The four elements are:

- (1). Hardware.
- (2). Software.
- (3). Liveware (Computer user).

**HARDWARE.**

Hardware is a term used to describe all the physical & tangible devices that make up a computer system, i.e. it consists of the parts that can be touched and felt.

Hardware include all mechanical & electronic elements found in the computer, e.g., the System Unit, Transistors, Diodes, bus systems such as electronic paths (channels), the Input devices (e.g., Keyboard, Mouse), Output devices (e.g., Monitor) & the Storage devices.

Hardware devices enable the user to enter information into a computer, view the output on screen, print out our work, store and process the work.

The hardware elements of a computer are generally grouped/ sub-divided into 4 major categories:-

1). Input devices.

Input devices are used to communicate with a computer. They enable the computer user to enter data, information & programs into the computer. They also let the user issue commands to the computer.

An Input device converts the input information into machine-sensible/ readable form.

Examples.

- * Keyboard.
- * Key-to-disk.
- * Scanner.
- * Trackball
- * Graphics pads (Tablets).
- * Speech Recognition devices.
- * Voice input devices, e.g. Microphones.
- * Document readers, such as, *Magnetic Ink Character Reader (MICR)*, *Optical Mark Reader (OMR)* & *Optical Character Reader (OCR)*.
- * Point Of Sale terminals, such as, *Bar code readers*, *Kimball Tag readers*, *Card readers*, & *Badge readers*.
- * Mouse.
- * Key-to-Tape.
- * Light pen.
- * Video digitizers.
- * Joystick / Game paddles.
- * Digital & Web cameras.

2). Central Processing unit – CPU (Processor).

The CPU is composed of the *Main Memory*, the *ALU* & the *Control unit*.

The CPU performs the necessary operations on the data held within the memory. It interprets & processes all the instructions from the Input devices.

The CPU is housed in the computer casing (**System Unit**), which contains all the major components of a computer system.

3). Output devices.

Output devices are used to extract/ disseminate processed data from the computer. They display the results of all the information that has been processed.

They also convert machine-coded output results from the Processor into a form that can be understood by people.

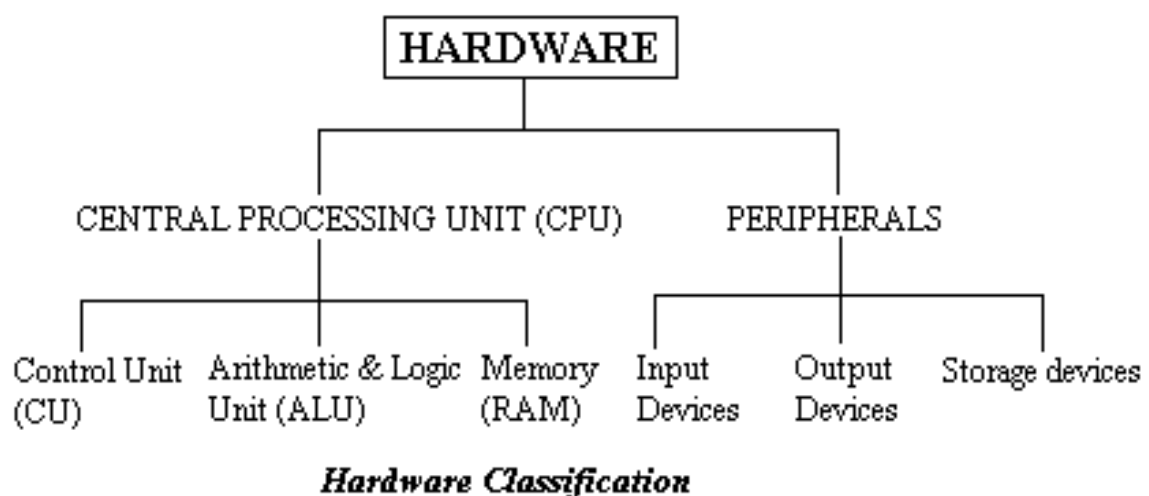
Examples.

- * Screen (Monitor/ Visual Display unit – VDU).
- * Audio Response units.
- * Sound output devices, e.g. Speakers.
- * Printers
- * Graph Plotters.
- * Microforms.

4). Storage devices.

These are devices used to store data & programs in computers. They include; Hard disks, Floppy disks, Magnetic tape drives, Cassette Tapes, Optical disks (CD-ROMs), and Random Access Memory (RAM).

Note. All these storage devices differ in the way (*technology*) they store data & the *capacities of data they can hold*.



Characteristics of Computer Hardware.

1. Hardware consists of parts that one can touch and feel.
2. Hardware determines what software will be used in the computer.
3. Computer hardware is expensive to acquire.
4. Hardware devices can only be made by specialist hardware engineers.
5. Not easy to change particular hardware components.

SOFTWARE.

- ❖ These are the programs & data used in a computer system that enable it perform a no. of specific functions.
- ❖ **Software** is a set of computer programs that guides the computer in each and every activity that happens inside the computer during data processing operations.

Software also includes the associated documentation (descriptions of the programs).

When used in a computer, Software instructs the computer to carry out specific processing tasks, e.g. produce the year end Balance sheet.

Characteristics of Computer Software.

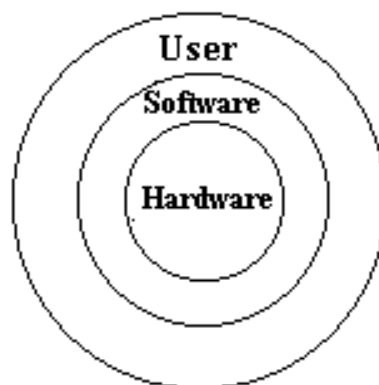
1. They are the programs & data used in a computer system.
2. A **Computer Program** is usually a set of computer instructions written in any of the computer programming languages, e.g. BASIC, PASCAL, etc.
3. It is not possible to see a program in memory as it exists in magnetic spots, however, you can see & touch a listing of the program on the computer screen.
4. Software enable computer hardware to operate effectively. In other words, software is meant to put 'life' into the hardware.
5. Software is flexible, i.e., the software used in a particular computer is relatively easy to change.
6. Software is cheaper compared to hardware devices.
7. Computer software can be written by the user, a Programmer or a Software house.

LIVEWARE

Liveware is a term used to refer to the computer end-user. They are the people who coordinate the various activities, which are necessary to get a computer system to perform useful tasks.

They include; Data entry operators, Computer Operators, Programmers, System Analysts, Data Processing Managers, Database Administrators, Computer Librarians, and the other staff directly or indirectly involved in the running of the system operations.

Apart from the hardware and software elements, the user is also seen as an integral part of the computer system as shown in the figure below;



LIVEWARE (Peopleware).

These are the people who coordinate the various activities, which are necessary to get a computer system to perform useful tasks.

Note that, a computer, left to itself, can do nothing.

Peopleware refers to all the people who are either working within the computer industry or are connected with it.

It includes all the people engaged in the manufacture (development) of computers or its components, maintenance and operations of computers. Such people may be working either in the software area or the hardware area or in both areas.

CLASSES OF END-USERS.

End-users are classified according to the kinds of software they use.

(a). Application end-user.

This is an end-user who puts the computer to some specific practical purpose by using an Application program.

Note. Application end-users are not usually technically knowledgeable about the computer, but may be trained to use a particular Application program.

(b). Computer specialist end-user.

This is an end-user whose work is to set-up, control & monitor the computer, and also produce or develop new software.

Note. Computer specialist end-users usually have technical knowledge about the computer, and therefore can control its operations.

Examples: Programmers, Computer operators, System managers, etc.

The following are some categories of Personnel normally employed in a computer department:

1. Data entry operators.

Are the people responsible for entering data (on disks or cards), to be processed by the computer.

2. Computer operators.

Are the people responsible for the day-to-day operation of the computer system.

3. Computer Programmers.

These are the people who write computer programs; which make the computer useful.

Duties.

- (i). Choosing of a Programming language to be used.
- (ii). Choosing of System software to be used.
- (iii). Testing of programs & giving the procedures to be used.
- (iv). Ensuring that Application programs are properly maintained.

4. System Analysts.

They design, and sometimes implement, the software systems required by the users. They usually work in close collaboration with the Programmers.

Duties.

- (i). Analyses the programs following the procedures given by the Programmer.
- (ii). They code and test all the programs used in the department level.

5. Data Processing Manager.

Is the person responsible for the overall management of the computer system.

Duties.

- (i). Responsible for all the services carried out in the department.
- (ii). Ensures that all the departments are efficient for the well-being of the company.
- (iii). He is the administrator.

Note. Hardware, Software & Liveware (people) are all important to the smooth operation of a computer system. A weakness in any the three will result in a weakness of the entire system.

This means that, the best written program may be useless if the hardware cannot handle it, and the fastest computer may be under-utilized if the software is not sophisticated enough. Lastly, the best machines with the best programs may be unproductive if the facilities are mismanaged or if careless data entry operators keep on entering incorrect data.

Exercise.

1. (a). Differentiate between an Application end-user and a Computer specialist.
(b). Briefly describe FIVE categories of workers which are normally found in commercial computer installation.

Exercise I.

1. (a). What is a computer system?
(b). State and briefly explain the THREE major elements of a computer system.
2. Differentiate between a Computer and a Computer system.
3. Explain briefly the meaning of the following computing terminologies:
 - (a). Microcomputer.
 - (b). Microcomputer system.
 - (c). Microprocessor system.
 - (d). Chip.
 - (e). Hardware.
 - (f). Software.
4. (a). Distinguish between 'Hardware' and 'Software' as used in a computer system.
(b). Other than output devices, name three other categories of computer hardware.

Exercise II.

1. (a). What is meant by a computer system?
(b). Identify and briefly describe each of the elements of a computer system.

ELECTRONIC COMPONENTS OF A MICROCOMPUTER SYSTEM.

A microcomputer consists of 4 electronic parts:

- 1). Input devices.
- 2). Central Processing Unit (CPU), also called the **Processor**.
- 3). Output devices.
- 4). Memory storage devices, which consist of Main memories & Secondary memories.

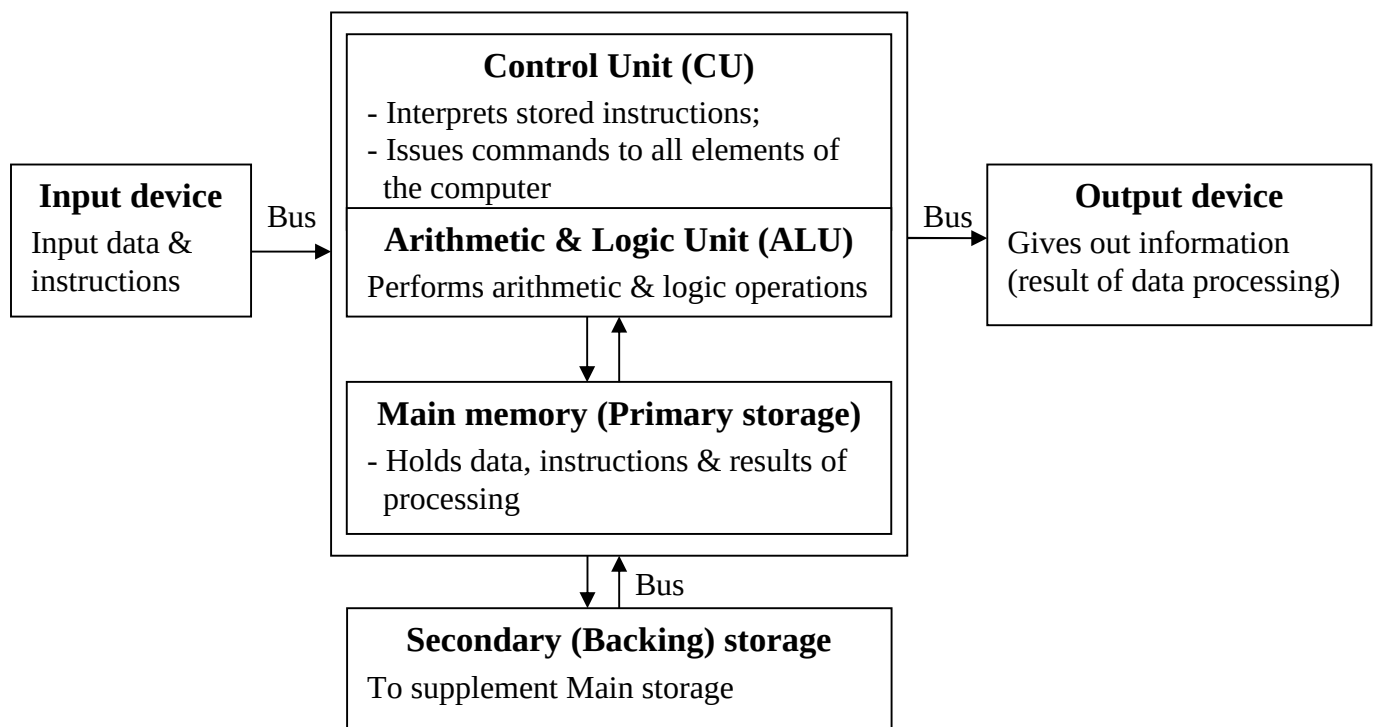


Fig. 1.2: A computer model

- ☞ Data & instructions to be processed are supplied to the computer memory by the user with the help of Input devices.
- ☞ The CPU performs the desired operations on the data and the results of calculations/processing are communicated to the user through the Output devices.
- ☞ The data and/or instructions not being used immediately by the computer are held permanently in the Backing storage, for retrieval any time it is required by the user.